

ARCRELAY

The Gasless Economic Layer for Arc

Enterprise Account Abstraction (ERC-4337) & Paymaster Infrastructure for Mainstream Value Movement

Document Type: Strategic Business Plan & Partner Pitch

Target Ecosystem: Arc Mainnet (Circle Stack / StableFX / Native USDC Gas)

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Executive Summary: ArcRelay is a B2B Account Abstraction infrastructure provider engineered natively for Arc. By leveraging Arc's native USDC gas token and deterministic sub-second finality, ArcRelay enables neobanks, institutional fintechs, and autonomous AI agents to eliminate gas UX friction entirely. We provide audited Smart Paymaster contracts, a high-throughput async bundler network, and enterprise sponsorship APIs that allow apps to pay transaction fees cleanly from standard USD balances.

1. The Core Problem: The Web3 UX Cliff

Every blockchain application faces an existential conversion hurdle: **the gas onboarding barrier**. Data across leading chains reveals that over **82% of non-crypto users drop off** at the moment they are asked to purchase and hold a volatile native token (like ETH, SOL, or POL) simply to pay transaction fees.

Why ERC-4337 is Broken on Legacy EVM Chains:

- **Dual-Asset Balance Sheet Risk:** Enterprise treasuries and fintechs cannot hold volatile crypto assets. On Ethereum or Arbitrum, sponsoring gas forces corporate clients to hold fluctuating ETH on their balance sheets.
- **Oracle Latency & Slippage:** Sponsoring gas in USDC on legacy chains requires continuous price oracle lookups (Chainlink/Pyth). Market volatility between UserOp submission and block inclusion causes failed transactions or severe spread losses.
- **DEX Swapping Friction:** Legacy paymasters collect USDC but must swap it for ETH on Uniswap/Curve to replenish their EntryPoint deposits, creating constant DEX fee leakage and MEV vulnerability.

The Arc Paradigm Shift:

Arc removes this entire friction layer at the protocol level. On Arc, **USDC IS the native gas token**. There are no volatile gas tokens, no DEX conversions, and no oracle lookups. A \$0.005 gas fee is paid directly with \$0.005 of native USDC. This transforms Account Abstraction from a fragile cryptographic hack into standard enterprise payment rails.

2. The ArcRelay Platform Architecture

ArcRelay is the institutional middleware connecting Arc's native protocol capabilities to client applications through a robust three-pillar architecture:

Pillar	Component	Core Function & Value
1. Smart Contract Layer	ArcRelay Smart Paymaster	Audited ERC-4337 Paymaster contract deployed on Arc. Holds client USDC tanks directly at the Arc EntryPoint. Validates user transactions via EIP-712 off-chain cryptographic signatures.
2. Off-Chain Bundler	Async Bundler & Policy Engine	High-throughput asynchronous Python/Rust daemon. Receives UserOps via JSON-RPC, enforces developer sponsorship rules (spend caps, rate limits), and batches executions directly to Arc validators.
3. B2B Developer Portal	ArcRelay Cloud Console & API	Web dashboard and API key manager. Enables developers to pre-fund corporate gas tanks via Circle Gateway, set fraud prevention rules, and monitor real-time user retention metrics.

3. Target Market & Customer Segments

Arc launched with over **190+ institutional and ecosystem builders** across on-chain FX, real-world assets, and cross-border payments. ArcRelay targets four primary verticals:

Vertical	Target Customer	Pain Point & ArcRelay Impact
Global Fintech & Payroll	Cross-border payment apps, remittance operators, contractor payout platforms.	Cannot expose end-users to blockchain gas mechanics. ArcRelay allows them to remit payroll with 100% predictable USD transaction deductions.
Autonomous AI Agents	Agentic economic workflows, data scraping bots, M2M compute settlement.	Agents need headless, autonomous execution without human wallet approval. ArcRelay provides zero-gas account abstraction for continuous micro-settlements.
On-Chain FX & Trading	StableFX market makers, DEX aggregators, liquidity pools.	High transaction velocity requires sub-second inclusion without front-running MEV. ArcRelay bundlers guarantee deterministic batch execution.
Tokenized RWA Platforms	Institutional bond, real estate, and treasury tokenization protocols.	Requires strict compliance and zero balance-sheet crypto volatility. ArcRelay provides audited corporate billing in pure fiat-backed USDC.

4. Business Model & Revenue Mechanics

ArcRelay employs a high-margin, scalable **infrastructure utility model** combining transaction volume markups with recurring enterprise SaaS licensing:

Revenue Stream 1: Sponsored Gas Markup (Volume Fee)

When a client app sponsors a user transaction, ArcRelay executes the UserOp and debits the corporate tank the exact Arc native gas fee plus a **5.0% – 8.5% platform convenience fee**. Because native gas fees on Arc are denominated in USDC, our margin is completely immune to token volatility.

Revenue Stream 2: B2B Enterprise SaaS Subscriptions

- **Starter (\$0/mo):** 8.5% gas markup, shared public bundler, up to \$2,500/mo gas spend.
- **Growth (\$349/mo):** 6.0% gas markup, priority bundler lane, custom policy rules, up to \$25,000/mo gas spend.
- **Enterprise (\$1,199/mo):** 4.5% gas markup, dedicated private bundler node, 99.99% uptime SLA, Circle Gateway auto-topup.

Metric / Year	Year 1 (Genesis)	Year 2 (Expansion)	Year 3 (Scale)
Active Client Applications	25 apps	110 apps	340 apps
Monthly Sponsored Gas Volume	\$250,000 / mo	\$1,500,000 / mo	\$5,200,000 / mo
Annual Gas Markup Revenue (6%)	\$180,000	\$1,080,000	\$3,744,000

Annual SaaS Subscription Revenue	\$105,000	\$462,000	\$1,428,000
Total Projected Annual Revenue	\$285,000	\$1,542,000	\$5,172,000

5. Phased Execution Roadmap

ArcRelay is structured for rapid, disciplined execution across four operational phases:

Phase 1: Core Engine & Smart Contract MVP (Months 1–2)

- Develop & verify the ArcRelay Paymaster Solidity contract conforming to ERC-4337 v0.7.
- Integrate native USDC gas handling and EIP-712 cryptographic signature verification.
- Deploy async Python bundler prototype with automated unit tests and load benchmarks.
- Deploy on Arc Testnet; conduct initial third-party security audit.

Phase 2: Developer Suite & Circle Stack Integration (Months 3–4)

- Build the ArcRelay Cloud Developer Console for API key and sponsorship policy configuration.
- Release lightweight TypeScript and Python client SDKs for seamless one-line app integration.
- Integrate Circle Gateway and CCTP for automated one-click corporate gas tank funding.
- Conduct closed beta with 3–5 initial ecosystem partners on Arc.

Phase 3: Mainnet Launch & Ecosystem Pilot (Months 5–6)

- Deploy production-grade multi-region Bundler cluster on Arc Mainnet.
- Launch commercial onboarding for Arc's 190+ launch partners.
- Implement enterprise monitoring, automated alerts, and real-time gas analytics.
- Achieve \$250k monthly sponsored volume threshold.

Phase 4: Institutional Scale & Advanced Agent Rails (Months 7+)

- Roll out specialized Circle Agent Stack hooks for autonomous AI micro-billing.
- Introduce KYB/AML screening integration for enterprise institutional clients.
- Expand cross-chain intent relaying connecting Arc liquidity to Ethereum, Arbitrum, and Base.

6. Partnership & Strategic Alignment

Why Arc and Circle Ecosystem Funds Should Back ArcRelay:

ArcRelay directly accelerates Arc's primary growth KPI: **daily active transactions and locked USDC velocity**. By removing UX friction for every dApp, game, fintech, and AI agent building on Arc, ArcRelay acts as a force multiplier for the entire ecosystem.

Immediate Next Steps:

1. Finalize technical specification and smart contract security scope.
2. Submit formal Ecosystem Grant Proposal to Arc Ecosystem Fund & Circle Ventures.
3. Initiate pilot integrations with the first 5 Arc launch dApps.

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